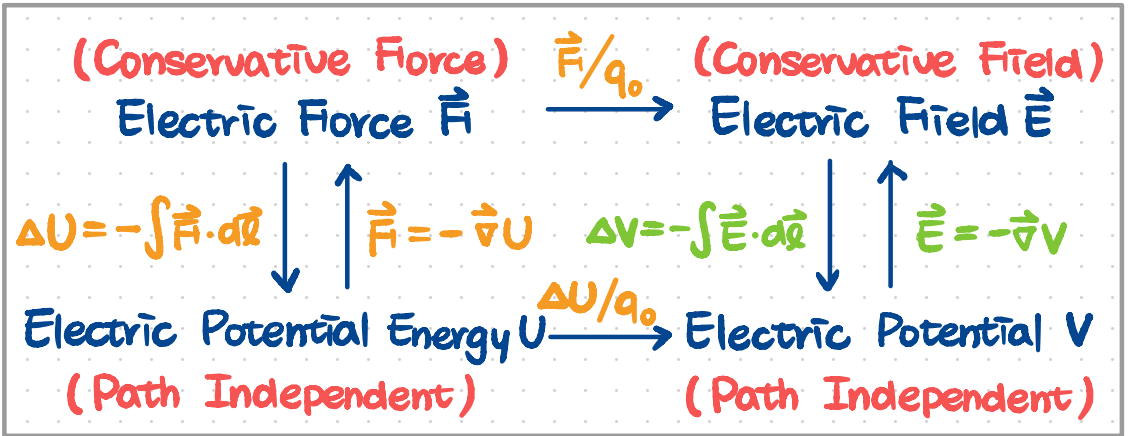


ILOs

1. $V_f - V_i = \Delta V \equiv - \int_i^f \vec{E} \cdot d\vec{r} \Leftrightarrow \vec{E} = - \vec{\nabla} V$
2. Distinguish between electric potential and electric potential energy.
3. Compute the electric potential and electric potential energy due to point charges and continuous charge distributions.

Ch23 Electric Potential 電位



$$V_f - V_i = \Delta V = - \int_i^f \frac{\vec{E} \cdot d\vec{r}}{dV} \quad \text{SI unit: J/C} = V$$

$$\Rightarrow dV = -\vec{E} \cdot d\vec{r} = -E_x dx - E_y dy - E_z dz$$

$$\Rightarrow \begin{cases} E_x = - \left. \frac{dV}{dx} \right|_{y,z=\text{const}} = - \frac{\partial V}{\partial x} \\ E_y = - \left. \frac{dV}{dy} \right|_{x,z=\text{const}} = - \frac{\partial V}{\partial y} \\ E_z = - \left. \frac{dV}{dz} \right|_{x,y=\text{const}} = - \frac{\partial V}{\partial z} \end{cases}$$

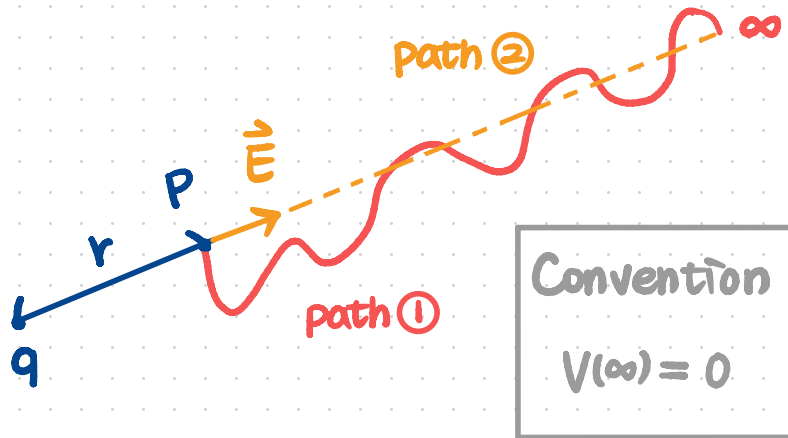
$$\Rightarrow \vec{E} = - \frac{\partial V}{\partial x} \hat{i} - \frac{\partial V}{\partial y} \hat{j} - \frac{\partial V}{\partial z} \hat{k} = -\nabla V$$

amount of electric potential energy added to the system when a test charge is brought from i to f .

$$\Delta U = q_0 \Delta V = \text{W}_{\text{other}}$$

the work you need to do to bring charge q_0 from i to f

Example: Point Charge



$$V_f - V_i = \Delta V = - \int_i^f \vec{E} \cdot d\vec{x}$$

$$V(P) - \underbrace{V(\infty)}_0 = - \int_{\infty}^P \vec{E} \cdot d\vec{x}$$

where

$$\vec{E} = \frac{q}{4\pi\epsilon_0 r'^2} \hat{r}'$$

$$d\vec{x} = dr' \hat{r}' \quad \text{Path ②}$$

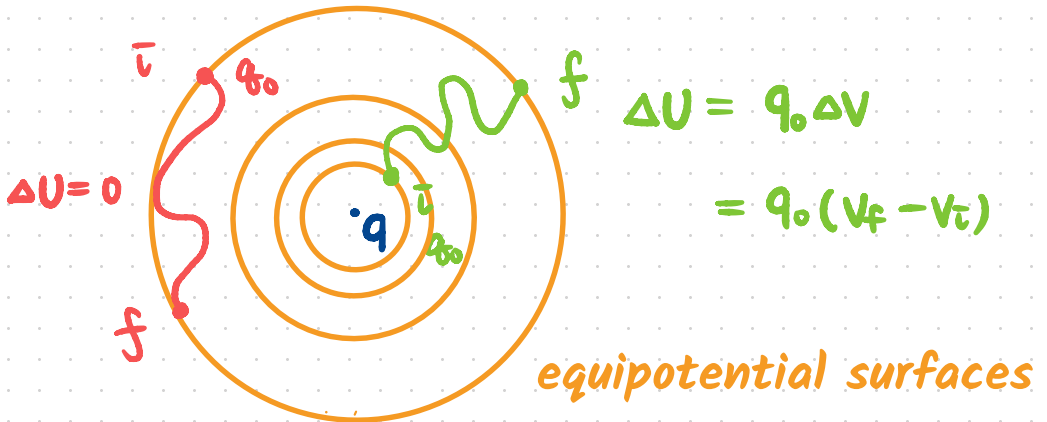
$$\Rightarrow V(r) = - \frac{q}{4\pi\epsilon_0} \int_{\infty}^r \frac{1}{r'^2} dr'$$

$$= + \frac{q}{4\pi\epsilon_0} \left(+ \frac{1}{r'} \right) \Big|_{\infty}^r$$

$$\boxed{V(r) = \frac{q}{4\pi\epsilon_0 r}} \quad \propto \frac{1}{r} \quad \begin{matrix} > 0 \\ = 0 \\ < 0 \end{matrix}$$

$$\vec{E} = - \vec{\nabla} V = - \frac{\partial V}{\partial r} \hat{r} = \frac{q}{4\pi\epsilon_0 r^2} \hat{r}$$

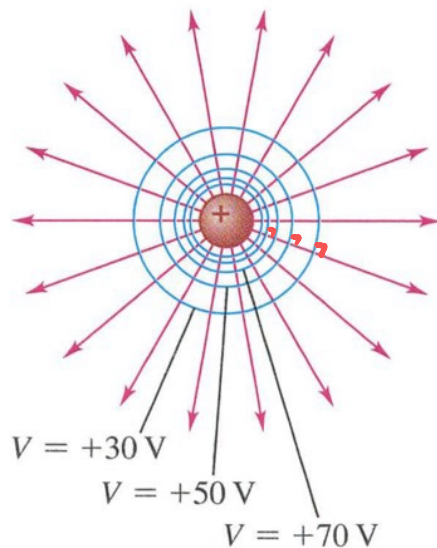
$$\text{Note: } \vec{\nabla} f(r, \theta, \phi) = \frac{\partial f}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial f}{\partial \theta} \hat{\theta} + \frac{1}{r \sin \theta} \frac{\partial f}{\partial \phi} \hat{\phi}$$



Equipotential Surface: a three-dimensional surface on which the electric potential V is the same at every point

Examples :

(a) A single positive charge

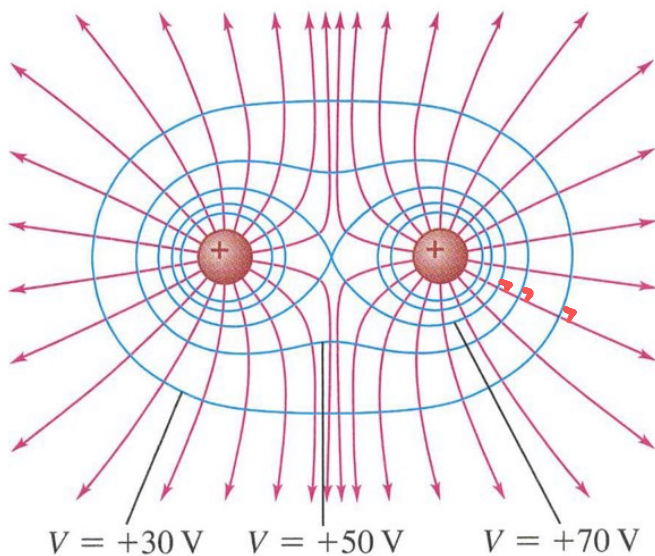


→ Electric field lines

Copyright © 2020 Pearson Education Ltd. All Rights Reserved

— Cross sections of equipotential surfaces

(c) Two equal positive charges



→ Electric field lines

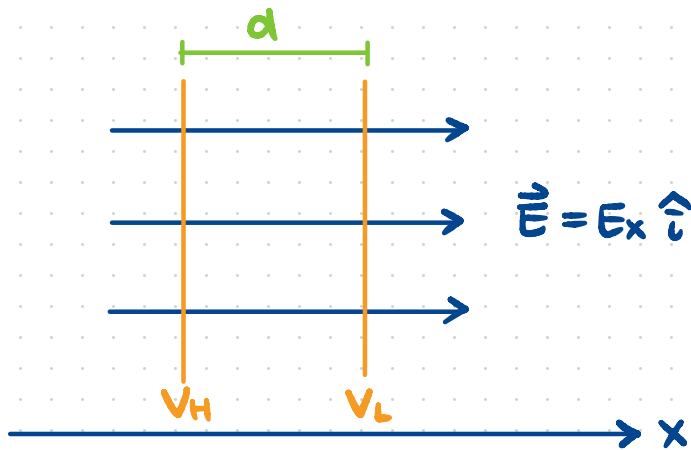
— Cross sections of equipotential surfaces

1. Electric field vector points from higher potential toward lower potential.

2. $\vec{E} \perp$ Equipotential Surfaces $\because \Delta V = -\int \vec{E} \cdot d\vec{r}$
 $\because \vec{E} \perp d\vec{r}$ when $\Delta V = 0$

3. In regions where the magnitude of $\vec{E}$ is large, the equipotential surfaces are close together because the field does a relatively large amount of work on a test charge in a relatively small displacement.

Example: Uniform Electric Field



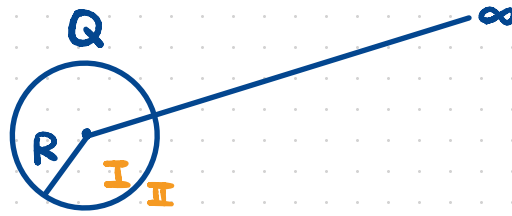
$$\Delta V = -\int \vec{E} \cdot d\vec{r} = -\int E_x dx = -E_x \Delta x$$

$$\Rightarrow V_H - V_L = E d$$

Example 23.8

Copyright © 2020 Pearson Education Ltd. All Rights Reserved

A solid conducting sphere of radius R has a total charge Q . Find the electric potential everywhere, both outside and inside the sphere.



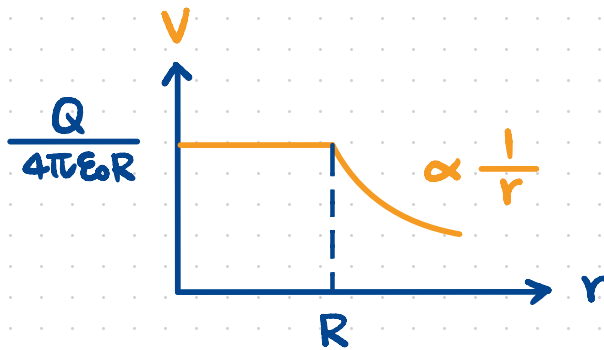
$$\vec{E} = \begin{cases} \vec{0} & \text{when } r < R \\ \frac{Q}{4\pi\epsilon_0 r^2} \hat{r} & \text{when } r > R \end{cases}$$

(II) $r > R$

$$V = - \int_{\infty}^r \frac{Q}{4\pi\epsilon_0 r'^2} dr' = \frac{Q}{4\pi\epsilon_0 r} \#$$

(I) $r < R$

$$\begin{aligned} V &= - \int_{\infty}^R \frac{Q}{4\pi\epsilon_0 r'^2} dr' - \int_R^r 0 dr' \\ &= \frac{Q}{4\pi\epsilon_0 R} \end{aligned}$$



Example 23.10 Copyright © 2020 Pearson Education Ltd. All Rights Reserved

Find the potential at a distance r from a very long line of charge with linear charge density λ .

A diagram showing a vertical line of charge. A horizontal distance r is marked from the line to a point. An electric field vector $\vec{E}$ is shown pointing away from the line.

$$\Delta V = - \int_i^f \vec{E} \cdot d\vec{x}$$

$$\Rightarrow V(r) - \underbrace{V(r_0)}_0 = - \int_{r_0}^r E_{r'} dr'$$

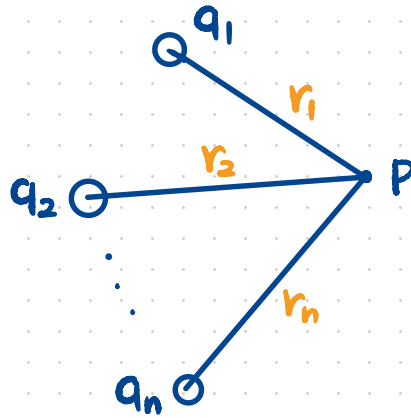
$$\Rightarrow V(r) = - \frac{\lambda}{2\pi\epsilon_0} \int_{r_0}^r \frac{dr'}{r'}$$

$$= \frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{r_0}{r}\right) *$$

$$\vec{E} = E_r \hat{r} = \frac{\lambda}{2\pi\epsilon_0 r} \hat{r}$$

We can not set $V(\infty) = 0$ when the charge distribution itself extends to infinity.

Electric Potential due to a Multi-Particle System



$$\Delta V = - \int_{\tau}^f \vec{E} \cdot d\vec{r}$$

$$\Rightarrow V_P - \frac{V_{\infty}}{0} = - \int_{\infty}^P \vec{E} \cdot d\vec{r}$$

where $\vec{E} = \vec{E}_1 + \vec{E}_2 + \dots + \vec{E}_n$ Principle of Superposition

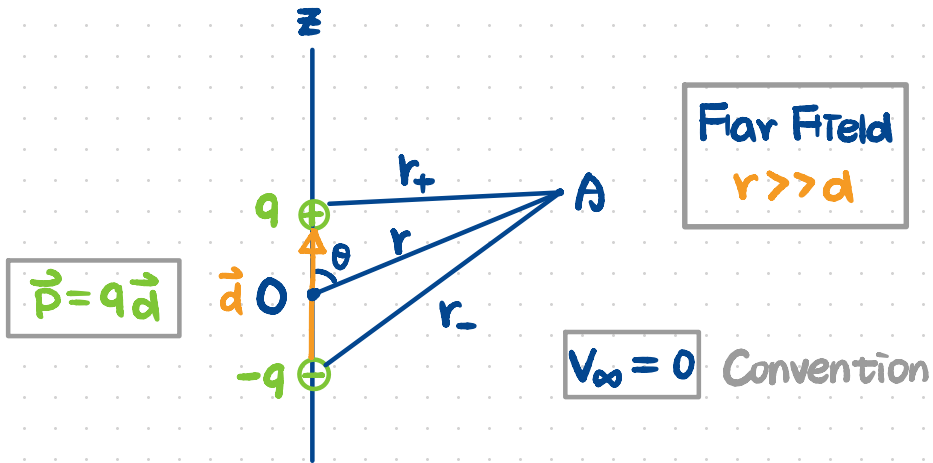
$$\Rightarrow V_P = - \int_{\infty}^P \vec{E}_1 \cdot d\vec{r} - \int_{\infty}^P \vec{E}_2 \cdot d\vec{r} \dots - \int_{\infty}^P \vec{E}_n \cdot d\vec{r}$$

$$= V_{P,1} + V_{P,2} + \dots + V_{P,n}$$

$$= \frac{q_1}{4\pi\epsilon_0 r_1} + \frac{q_2}{4\pi\epsilon_0 r_2} + \dots + \frac{q_n}{4\pi\epsilon_0 r_n}$$

$$\Rightarrow \boxed{V_P = \frac{1}{4\pi\epsilon_0} \sum_i \frac{q_i}{r_i}} \quad (V_{\infty} = 0)$$

Example: Electric Dipole



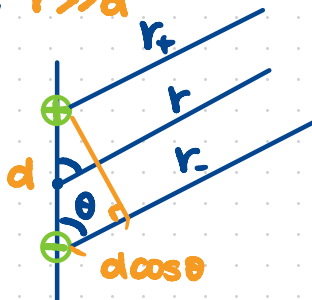
$$V_A = \frac{+q}{4\pi\epsilon_0 r_+} + \frac{-q}{4\pi\epsilon_0 r_-}$$

$$= \frac{q}{4\pi\epsilon_0} \left(\frac{1}{r_+} - \frac{1}{r_-} \right)$$

$$= \frac{q}{4\pi\epsilon_0} \frac{r_- - r_+}{r_+ r_-}$$

$$\stackrel{r \gg d}{=} \frac{q d \cos \theta}{4\pi\epsilon_0 r^2} = \frac{\vec{p} \cdot \hat{r}}{4\pi\epsilon_0 r^2} \propto \frac{1}{r^2} *$$

Note that $r \gg d$

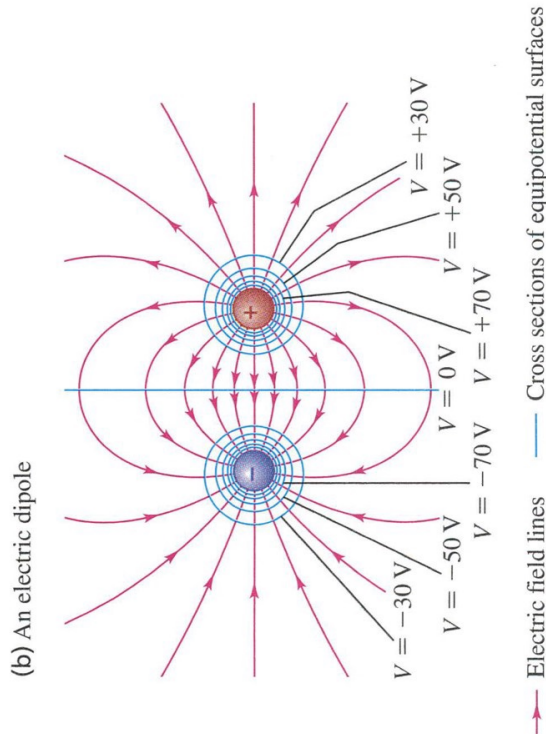


$$r_+ r_- \approx r^2$$

$$r_- - r_+ \approx d \cos \theta$$

$$\vec{E}_A = -\vec{\nabla}V = \frac{3(\vec{p} \cdot \hat{r})\hat{r} - \vec{p}}{4\pi\epsilon_0 r^3} \propto \frac{1}{r^3}$$

Note: $\vec{\nabla}f(r, \theta, \phi) = \frac{\partial f}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial f}{\partial \theta} \hat{\theta} + \frac{1}{r \sin \theta} \frac{\partial f}{\partial \phi} \hat{\phi}$



Copyright © 2020 Pearson Education Ltd. All Rights Reserved

Electric Potential due to Continuous Charges

$$V = \int dV = \frac{1}{4\pi\epsilon_0} \int \frac{dq}{r} \quad (V_\infty = 0)$$

Line Charge : $dq = \lambda dl$

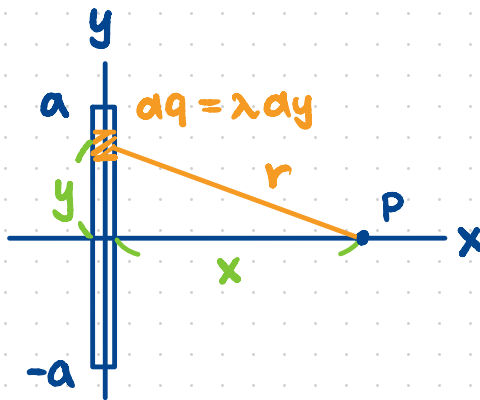
Surface Charge : $dq = \sigma dA$

Volume Charge : $dq = \rho dV$

Example 23.12

Copyright © 2020 Pearson Education Ltd. All Rights Reserved

Positive electric charge Q is distributed uniformly along a line of length $2a$ lying along the y -axis between $y = -a$ and $y = +a$. Find the electric potential at a point P on the x -axis at a distance x from the origin.



$$dq = \lambda dy$$
$$\lambda = \frac{Q}{2a}$$

$$dV = \frac{1}{4\pi\epsilon_0} \frac{dq}{r} = \frac{1}{4\pi\epsilon_0} \frac{\lambda dy}{\sqrt{x^2 + y^2}}$$

$$\begin{aligned}
V &= \int dV \\
&= \frac{\lambda}{4\pi\epsilon_0} \int_{-a}^a \frac{dy}{\sqrt{x^2+y^2}} \\
&= \frac{\lambda}{4\pi\epsilon_0} \ln(\sqrt{x^2+y^2} + y) \Big|_{-a}^a \\
&= \frac{1}{4\pi\epsilon_0} \frac{Q}{2a} \ln \left(\frac{\sqrt{x^2+a^2} + a}{\sqrt{x^2+a^2} - a} \right) \quad \text{✱}
\end{aligned}$$

Note that $\int \frac{dy}{\sqrt{x^2+y^2}}$

$$= \int \sec\theta \, d\theta$$

$$\begin{aligned}
y &= x \tan\theta \\
dy &= x \sec^2\theta \, d\theta
\end{aligned}$$

$$\begin{aligned}
&= \int \sec\theta \frac{\sec\theta + \tan\theta}{\sec\theta + \tan\theta} \, d\theta \\
&= \int \frac{du}{u}
\end{aligned}$$

$$\begin{aligned}
u &= \sec\theta + \tan\theta \\
du &= (\sec\theta \tan\theta + \sec^2\theta) \, d\theta
\end{aligned}$$

$$\begin{aligned}
&= \ln|u| + \text{const} \\
&= \ln|\sec\theta + \tan\theta| + \text{const} \\
&= \ln \left| \frac{\sqrt{x^2+y^2}}{x} + \frac{y}{x} \right| + \text{const} \\
&= \ln |\sqrt{x^2+y^2} + y| + \text{const}
\end{aligned}$$

$$E_x = - \frac{\partial V}{\partial x}$$

$$= \frac{-1}{4\pi\epsilon_0} \frac{Q}{2a} \left[\frac{\frac{1}{2}(x^2+a^2)^{-\frac{1}{2}} \cdot 2x}{\sqrt{x^2+a^2}+a} - \frac{\frac{1}{2}(x^2+a^2)^{-\frac{1}{2}} \cdot 2x}{\sqrt{x^2+a^2}-a} \right]$$

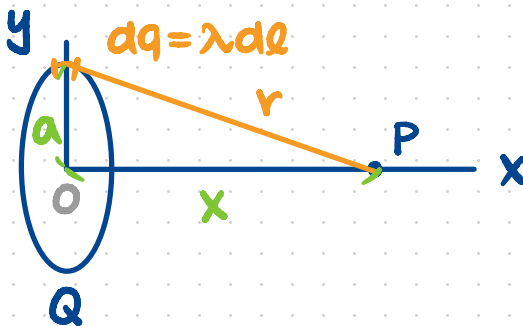
$$= \frac{-1}{4\pi\epsilon_0} \frac{Q}{2a} \frac{x}{\sqrt{x^2+a^2}} \left[\frac{\sqrt{x^2+a^2}-a - (\sqrt{x^2+a^2}+a)}{x^2+a^2-a^2} \right]$$

$$= \frac{1}{4\pi\epsilon_0} \frac{Q}{x\sqrt{x^2+a^2}} *$$

Example 23.11

Copyright © 2020 Pearson Education Ltd. All Rights Reserved

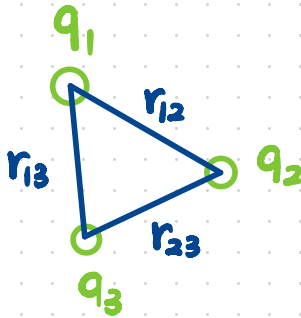
Electric charge Q is distributed uniformly around a thin ring of radius a . Find the potential and the electric field at a point P on the ring axis at a distance x from the center of the ring.



Electric Potential Energy

The **total work** required to assemble the charges from **infinity** to their current configuration.

e.g. 3 point charges



$$U = q_2 V_{10} + q_3 V_{12}$$
$$= \frac{k q_1 q_2}{r_{12}} + \frac{k q_1 q_3}{r_{13}} + \frac{k q_2 q_3}{r_{23}}$$

The total potential energy of a system of particles is the sum of the potential energies for every pair of particles in the system

$$= \frac{1}{2} \sum_i q_i V_i$$

Example 23.1

Copyright © 2020 Pearson Education Ltd. All Rights Reserved

A positron has mass 9.11×10^{-31} kg and charge $q_0 = +e = +1.60 \times 10^{-19}$ C. Suppose a positron moves in the vicinity of an α particle, which has charge $q = +2e = +3.20 \times 10^{-19}$ C and mass 6.64×10^{-27} kg. The α particle's mass is more than 7000 times that of the positron, so we assume that the α particle remains at rest. When the positron is 1.00×10^{-10} m from the α particle, it is moving directly away from the α particle at 3.00×10^6 m/s.

(a) What is the positron's speed when the particles are 2.00×10^{-10} m apart?

Only electric force does work

$$\Rightarrow E_{\text{mec}} = \text{const}$$

$$E_{\text{mec},i} = E_{\text{mec},f}$$

$$U_i + K_i = U_f + K_f$$

$$\frac{kqq_0}{r_i} + \frac{1}{2} m_e v_i^2 = \frac{kqq_0}{r_f} + \frac{1}{2} m_e v_f^2$$

$$\Rightarrow K_f = \frac{1}{2} m_e v_f^2 = kqq_0 \left(\frac{1}{r_i} - \frac{1}{r_f} \right) + \frac{1}{2} m_e v_i^2$$

$$\begin{aligned}
 &= (9.0 \times 10^9) (3.20 \times 10^{-19}) (1.60 \times 10^{-19}) \\
 &\quad \times \left(\frac{1}{1.00 \times 10^{-10}} - \frac{1}{2.00 \times 10^{-10}} \right) \\
 &\quad + \frac{1}{2} \times 9.11 \times 10^{-31} \times (3.00 \times 10^6)^2 \\
 &= 6.40 \times 10^{-18} \text{ J}
 \end{aligned}$$

$$\Rightarrow v_f = \sqrt{\frac{2K_f}{m_e}} = 3.75 \times 10^6 \text{ m/s} \#$$

(b) What is the positron's speed when it is very far from the α particle?

$$E_{\text{mec},i} = E_{\text{mec},f}$$

$$U_i + K_i = U_f + K_f$$

$$\frac{kq q_0}{r_i} + \frac{1}{2} m_e v_i^2 = 0 + \frac{1}{2} m_e v_f^2$$

$$\Rightarrow K_f = \frac{1}{2} m_e v_f^2$$

$$= \frac{(9.0 \times 10^9) (3.20 \times 10^{-19}) (1.60 \times 10^{-19})}{1.00 \times 10^{-10}}$$

$$+ \frac{1}{2} \times 9.11 \times 10^{-31} \times (3.00 \times 10^6)^2$$

$$= 8.71 \times 10^{-18} \text{ J}$$

$$\Rightarrow v_f = \sqrt{\frac{2k_f}{m_e}} = 4.37 \times 10^6 \text{ m/s} *$$

(C) Suppose the initial conditions are the same but the moving particle is an electron. Describe the subsequent motion.

$$\begin{aligned} E_{\text{mec}, i} &= - \frac{k q e}{r_i} + \frac{1}{2} m_e v_i^2 \\ &= - \frac{(9.0 \times 10^9) (3.20 \times 10^{-19}) (1.60 \times 10^{-19})}{1.00 \times 10^{-10}} \\ &\quad + \frac{1}{2} \times 9.11 \times 10^{-31} \times (3.00 \times 10^6)^2 \\ &= - 5.09 \times 10^{-19} \text{ J} < 0 \end{aligned}$$

e would reach a maximum separation $r = r_m$ from the α particle and reverse its direction.

$$U_i + K_i = U_m + 0$$

$$\Rightarrow U_m = - \frac{k q e}{r_m} = - 5.09 \times 10^{-19} \text{ J}$$

$$\begin{aligned} \Rightarrow r_m &= - \frac{k q e}{U_m} = + \frac{(9.0 \times 10^9) (3.20 \times 10^{-19}) (1.60 \times 10^{-19})}{+ 5.09 \times 10^{-19}} \\ &= 9.05 \times 10^{-10} \text{ m} * \end{aligned}$$

Electrostatics

Integral Form $\Leftrightarrow$ Differential Form

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{encl}}}{\epsilon_0}$$

$$\oint \vec{E} \cdot d\vec{r} = 0$$

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

$$\vec{\nabla} \times \vec{E} = \vec{0}$$

Stokes' Theorem right-hand rule

$$\oint_{\partial} \vec{E} \cdot d\vec{r} = \int_S (\vec{\nabla} \times \vec{E}) \cdot d\vec{A}$$



$$\Rightarrow \int_S (\vec{\nabla} \times \vec{E}) \cdot d\vec{A} = 0$$

for an arbitrary surface

$$\Rightarrow \vec{\nabla} \times \vec{E} = \vec{0} \quad *$$

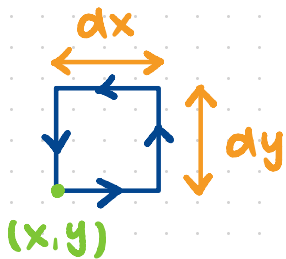
where

$$\vec{\nabla} \times \vec{E} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ E_x & E_y & E_z \end{vmatrix}$$

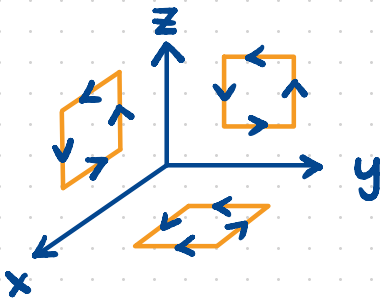
$$= \left(\frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z} \right) \hat{i} + \left(\frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x} \right) \hat{j} + \left(\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} \right) \hat{k}$$

proof of Stokes' Theorem

tiny square



$$\begin{aligned} \oint \vec{E} \cdot d\vec{r} \\ &\stackrel{HW}{=} \left(\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} \right) dx dy \\ &= (\vec{\nabla} \times \vec{E})_z dA_z \\ &= (\vec{\nabla} \times \vec{E}) \cdot d\vec{A} \end{aligned}$$



Any arbitrary loop can be composed of many tiny loops.

$$\Rightarrow \sum_i \oint_i \vec{E} \cdot d\vec{r} = \sum_i \lim_{\Delta A_i \rightarrow 0} (\vec{\nabla} \times \vec{E}) \cdot \Delta \vec{A}_i$$

$$\Rightarrow \oint_{\mathcal{L}} \vec{E} \cdot d\vec{r} = \int_S (\vec{\nabla} \times \vec{E}) \cdot d\vec{A}$$

Q.E.D.